

Practical-7

```
himasree@LAPTOP-41COT5AA:~/OSLab$ nano memory.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc memory.c -o memory
himasree@LAPTOP-41COT5AA:~/OSLab$ ./memory
===== PROCESS MEMORY ADDRESSES =====
```

```
Code address      : 0x61c88b805229
Global variable   : 0x61c88b808010
Static variable   : 0x61c88b808014
Local static      : 0x61c88b808018
Heap variable     : 0x61c8b8b3a2a0
Stack variable    : 0x7fff917cf85c
```

```
===== PROCESS INFORMATION =====
Process ID (PID)   : 1246
```

```
Keep this process running to inspect /proc/1246/maps
Press Enter to exit...
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cat /proc/1246/maps
61c88b804000-61c88b805000 r--p 00000000 08:30 13110 /home/himasree/OSLab/memory
61c88b805000-61c88b806000 r-xp 00001000 08:30 13110 /home/himasree/OSLab/memory
61c88b806000-61c88b807000 r--p 00002000 08:30 13110 /home/himasree/OSLab/memory
61c88b807000-61c88b808000 r--p 00002000 08:30 13110 /home/himasree/OSLab/memory
61c88b808000-61c88b809000 rw-p 00003000 08:30 13110 /home/himasree/OSLab/memory
61c8b8b3a000-61c8b8b5b000 rw-p 00000000 00:00 0 [heap]
79f920a00000-79f920a28000 r--p 00000000 08:30 44587 /usr/lib/x86_64-linux-gnu/libc.so.6
79f920a28000-79f920bb0000 r-xp 00028000 08:30 44587 /usr/lib/x86_64-linux-gnu/libc.so.6
79f920bb0000-79f920bfff000 r--p 001b0000 08:30 44587 /usr/lib/x86_64-linux-gnu/libc.so.6
79f920bfff000-79f920c03000 r--p 001fe000 08:30 44587 /usr/lib/x86_64-linux-gnu/libc.so.6
79f920c03000-79f920c05000 rw-p 00202000 08:30 44587 /usr/lib/x86_64-linux-gnu/libc.so.6
79f920c05000-79f920c12000 rw-p 00000000 00:00 0
79f920c12000-79f920cf0000 rw-p 00000000 00:00 0
79f920cf0000-79f920d02000 rw-p 00000000 00:00 0
79f920d02000-79f920d06000 r--p 00000000 00:00 0 [vvar]
79f920d06000-79f920d08000 r--p 00000000 00:00 0 [vvar_vclock]
79f920d08000-79f920d0a000 r-xp 00000000 00:00 0 [vdso]
79f920d0a000-79f920d0b000 r--p 00000000 08:30 44584 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
79f920d0b000-79f920d36000 r-xp 00001000 08:30 44584 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
79f920d36000-79f920d40000 r--p 0002c000 08:30 44584 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
79f920d40000-79f920d42000 r--p 00036000 08:30 44584 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
79f920d42000-79f920d44000 rw-p 00038000 08:30 44584 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7fff917b0000-7fff917d1000 rw-p 00000000 00:00 0 [stack]
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ grep heap /proc/1246/maps
61c8b8b3a000-61c8b8b5b000 rw-p 00000000 00:00 0 [heap]
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ grep stack /proc/1246/maps
7fff917b0000-7fff917d1000 rw-p 00000000 00:00 0 [stack]
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ grep memory /proc/1246/maps
61c88b804000-61c88b805000 r--p 00000000 08:30 13110 /home/himasree/OSLab/memory
61c88b805000-61c88b806000 r-xp 00001000 08:30 13110 /home/himasree/OSLab/memory
61c88b806000-61c88b807000 r--p 00002000 08:30 13110 /home/himasree/OSLab/memory
61c88b807000-61c88b808000 r--p 00002000 08:30 13110 /home/himasree/OSLab/memory
61c88b808000-61c88b809000 rw-p 00003000 08:30 13110 /home/himasree/OSLab/memory
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ pmap 1246
1246:      ./memory
000061c88b804000      4K r---- memory
000061c88b805000      4K r-x-- memory
000061c88b806000      4K r---- memory
000061c88b807000      4K r---- memory
000061c88b808000      4K rw--- memory
000061c8b8b3a000     132K rw--- [ anon ]
000079f920a00000     160K r---- libc.so.6
000079f920a28000    1568K r-x-- libc.so.6
000079f920bb0000     316K r---- libc.so.6
000079f920bfff00      16K r---- libc.so.6
000079f920c03000       8K rw--- libc.so.6
000079f920c05000      52K rw--- [ anon ]
000079f920cf6000      12K rw--- [ anon ]
000079f920d00000       8K rw--- [ anon ]
000079f920d02000      16K r---- [ anon ]
000079f920d06000       8K r---- [ anon ]
000079f920d08000       8K r-x-- [ anon ]
000079f920d0a000       4K r---- ld-linux-x86-64.so.2
000079f920d0b000     172K r-x-- ld-linux-x86-64.so.2
000079f920d36000      40K r---- ld-linux-x86-64.so.2
000079f920d40000       8K r---- ld-linux-x86-64.so.2
000079f920d42000       8K rw--- ld-linux-x86-64.so.2
00007fff917b0000     132K rw--- [ stack ]
total                2688K
himasree@LAPTOP-41COT5AA:~/OSLab$

```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ cat memory.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;          // Global variable
static int static_var = 20;   // Static variable

void code_function()
{
    printf("Code address      : %p\n", (void *)code_function);
}

int main()
{
    int stack_var = 30;        // Stack variable
    static int local_static = 40; // Static variable
    int *heap_var;

    // Allocate memory on heap
    heap_var = (int *)malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc failed");
        return 1;
    }

    *heap_var = 50;
}

```

```
*heap_var = 50;

printf("==== PROCESS MEMORY ADDRESSES =====\n\n");

printf("Code address      : %p\n", (void *)code_function);
printf("Global variable    : %p\n", (void *)&global_var);
printf("Static variable     : %p\n", (void *)&static_var);
printf("Local static        : %p\n", (void *)&local_static);
printf("Heap variable       : %p\n", (void *)heap_var);
printf("Stack variable      : %p\n", (void *)&stack_var);

printf("\n==== PROCESS INFORMATION =====\n");

printf("Process ID (PID)    : %d\n", getpid());

printf("\nKeep this process running to inspect /proc/%d/maps\n", getpid());

printf("Press Enter to exit...\n");
getchar();

free(heap_var);

return 0;
}
```